BSMS-Physics

SEMESTER - I						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA 101	Calculus	3	1.5	0	9
2	PH 101	Quantum Physics and Applications	2	1	0	6
3	**CH101T**	Fundamental Concepts and Applications of Chemistry	3	0	0	6
4	BB 103	Introduction to Modern biology	3	0	0	6
5	PH 113	Hands on Science Laboratory - I	0	0	3	3
6	CS 101	Computer Programming	3	0	3	9
7	HS 106	Design Thinking and Creativity	1	0	0	PP/NP
8	NO107/ NO105/	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					41

Semester - II						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS201	Data Structure & Algorithms	3	0	0	6
2	BB201	Biomolecules	2	1	0	6
3	MA XXX	Linea lgebra and Differential Equations	3	1.5	0	9
4	PH102	Electricity and Magnetism	2	1	0	6
5	**CH203T**	States of Matter	3	0	0	3
6	**CH201T**	Organic Chemisrty	3	0	0	3
7	CS211	Data Structure & Algorithms Lab	0	0	3	3
8	**CH101L**	Hands on Science Loboratory-II	0	0	3	3
9	NO107/ NO105	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				41

Semester III						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	HS 201	Economics	3	0	0	6
2	BB 301	Basics of Cell Biology and Genetics	3	0	0	6
3	**CH204T**	Physical Organic and Bioorganic Chemistry (1st Half)	3	0	0	3
4	**CH202T**	Inorganic Chemistry (2nd Half)	3	0	0	3
5	MA 209	Introduction to Probability Theory	3	1	0	8
6	PH 205	Waves, Oscillations & Optics	2	1	0	6
7	MA 223	Mathematics Laboratory	0	0	3	3
		Total Credits				35

Semester - IV						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE 301	Environmental Studies	3	0	0	6
2	EE 229	Electronic Device (Post Mid Sem)	3	0	0	3

3	PH 322	Mathematical Physics-I	2	1	0	6
4		Program Elective-I	2	1	0	6
5		HSS Elective-I	3	0	0	6
6	PH 312	General Physics Laboratory	0	0	3	3
		Total Credits				30

Semester V						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH301T	Electrodynamics	2	1	0	6
2	PH302T	Classical Mechanics	3	0	0	6
3	PH303T	Quantum Mechanics - I	2	1	0	6
4	PH420T	Electronics	2	1	0	6
5	PH501T	Mathematical Physics-II	2	1	0	6
6	PH402L	Electronics Laboratory	0	0	3	3
		Total Credits				33

Semester VI						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH401T	Quantum Mechanics - II	2	1	0	6
2	PH305T	Statistical Physics	3	0	0	6
3	PH402T	Condensed Matter Physics	2	1	0	6
4		Institute Elective-I	2	1	0	6
5		Program Elective-I	2	1	0	6
6		Institute Elective-II	2	1	0	6
		Total Credits				36

Semester VII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH501C	Numerical Methods	2	1	0	6
2	PH417T	Nuclear and Particle Physics	2	1	0	6
3	PH401L	Advanced Physics Laboratory	0	0	3	3
4		Program Elective-II	2	1	0	6
5		Institute Elective-III/PH501P:RnD Project	2	1	0	6
		Seminar-I	0	0	4	4
		Total Credits				31

Semester VIII						
S.No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	PH414T	Atomic and Molecular Physics	2	1	0	6
2		Institute Elective-IV	2	1	0	6
3		Institute Elective-V/PH502P R&D Project-2	2	1	0	6
4		Program Elective-III	2	1	0	6
		Total Credits				24

		Overal Credits Required (Minimum)	**271**
Semester IX & X			
		Independent Project	60
		Total Credits	60
		Overal Credits Required (Minimum)	**331**

BSMS-Mathematics

SEMESTER - I						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA108T	Calculus	3	1.5	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	CY101T	Fundamental Concepts and Applications of Chemistry	3	0	0	6
4	BB101T	Introduction to Modern biology	3	0	0	6
5	PH101L	Hands on Science Laboratory - I	0	0	3	3
6	CS101C	Computer Programming	3	0	3	9
7	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					41

Semester - II						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS101T	Data Structures and Algorithms	3	0	0	6
2	BB201T	Biomolecules	2	1	0	6
3	MA109T	Linea lgebra and Differential Equations	3	1.5	0	9
4	PH102T	Electricity and Magnetism	2	1	0	6
5	CY203T	States of Matter	3	0	0	3
6	CH201T	Organic Chemisrty	3	0	0	3
7	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
8	PH101L	Hands on Science Loboratory-II	0	0	3	3
9	NO107/ NO105/ NO	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				41

Semester III						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	HS201T	Economics	3	0	0	6
2	BB401T	Basics of Cell Biology and Genetics	3	0	0	6
3	CY204T	Physical Organic and Bioorganic Chemistry (1st Half)	3	0	0	3
4	CY202T	Inorganic Chemistry (2nd Half)	3	0	0	3
5	MA401T	Introduction to Probability Theory	3	1	0	8
6	PH201T	Waves, Oscillations & Optics	2	1	0	6
7	MA102L	Mathematics Laboratory	0	0	3	3
		Total Credits				35

Semester - IV						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	MA302T	Real Analysis	2	1	0	6
2	MA303T	Group Theory	2	1	0	6

3	MA107T	Advanced Linear Algebra	2	1	0	6
4		Program Elective-I				6
5		Program Elective-II				6
		Total Credits				30

Semester V

S.No	Course Code	Course Name	L	T	P	C
1	MA304T	Rings and Modules	2	1	0	6
2	MA405T	Introduction to Complex Analysis	2	1	0	6
3	MA406T	General Topology	2	1	0	6
4	MA414T	Graph Theory and Combinatorics	2	1	0	6
5		Program Elective-II				6
		Total Credits				30

Semester VI

S.No	Course Code	Course Name	L	T	P	C
1	CE301T	Environmental Studies	3	0	0	6
	MA403T	Ordinary Differential Equations	2	1	0	6
	MA602T	Measure Theory	2	1	0	6
		Program Elective-III				6
		HSS Elective-I	3	0	0	6
		Total Credits				30

Semester VII

S.No	Course Code	Course Name	L	T	P	C
1	MA412T	Numerical Analysis	2	1	0	6
2		Program Elective-IV				6
3		Program Elective-V				3
4		Institute Elective – I	2	1	0	6
5		HSS Elective-II	3	0	0	6
		Total Credits				27

Semester VIII

S.No	Course Code	Course Name	L	T	P	C
1		Program Elective-VI	2	1	0	6
2		Institute Elective – II	2	1	0	6
3		Institute Elective – III				6
4		Institute Elective – IV				6
5		Institute Elective – v				6
		Total Credits				30

Semester IX

1		**Independent Project**	**30**
Semester X			
2		**Independent Project**	**30**
		Total Credits	**60**
		Overal Credits Required (Minimum)	**321**

BSMS- Chemistry

SEMESTER - I (Common for all B.Tech Courses)

S.No	Course Code	Course	L	T	P	C
1	MA108T	Calculus	3	1.5	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	CY101T	Fundamental Concepts and Applications of Chemistry	3	0	0	6
4	BB101T	Introduction to Modern biology	3	0	0	6
5	PH101L	Hands on Science Laboratory - I	0	0	3	3
6	CS101C	Computer Programming	3	0	3	9
7	HS101T	Design Thinking and Creativity	1	0	0	PP/ NP
8	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
	Total credit					**41**

Semester - II

S. No	Course Code	Course Name	L	T	P	C
1	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
2	CS101T	Data Structures and Algorithms	3	0	0	6
3	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
4	PH102T	Electricity and Magnetism	2	1	0	6
5	BB201T	Biomolecules	2	1	0	6
6	CY203T	States of Matter (1st Half)	3	0	0	3
7	CH201T	Organic Chemistry (2nd Half)	3	0	0	3
8	PH101L	Hands On Science Laboratory - II	0	0	3	3
9	NO107/ NO105/ NO 106/ NO 108	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				41

Semester III

S. No	Course Code	Course Name	L	T	P	C
1	HS201T	Economics	3	0	0	6
2	BB401T	Basics of Cell Biology and Genetics	3	0	0	6
3	CY202T	Inorganic Chemistry (1st Half)	3	0	0	3
4	CY204T	Physical Organic and Bioorganic Chemistry (2nd Half)	3	0	0	3
5	MA401T	Introduction to Probability Theory	3	1	0	9
6	PH201T	Waves, Oscillations & Optics	2	1	0	6
7	MA102L	Mathematics Laboratory	0	0	3	3

	Total credit	**36**

Semester - IV

S. No	Course Code	Course Name	L	T	P	C
1	CE301T	Environmental studies	3	0	0	6
2	CY401T	Organic Reactions and Reagents	3	0	0	6
3	CY411T	Quantum Chemistry	3	0	0	6
4	CY402T	Molecular Energetics and Dynamics	3	0	0	6
5		Institute elective -I	3	0	0	6
6	CY301L	Chemistry laboratory-I	0	0	3	3
	Total credit					33

Semester V

S. No	Course Code	Course Name	L	T	P	C
1	CH512T	Transition Metals and Coordination Chemistry	3	0	0	6
2	CH513T	Concepts and Mechanismsin organic chemistry	3	0	0	6
3	CH503T	Molecular spectroscopy	3	0	0	6
4	CH511T	Main group chemistry	3	0	0	6
	CH401L	Chemistry laboratory-II	0	0	3	3
5		Program Elective-I	2	1	0	3
6		Program Elective-II	2	1	0	3
	Total credit					**33**

Semester VI

S. No	Course Code	Course Name	L	T	P	C
1	CH514T	Organometallic chemistry and catalysis	3	0	0	6
2	CH515T	Instrumental Methodsfor Structure Determination	3	0	0	6
3	CH402L	Chemistry laboratory-III	0	0	3	3
4		Program Elective-III	2	1	0	3
5		Program Elective-IV	2	1	0	6
6		Institute Elective – II	3	0	0	6
	Total credit					**30**

Semester VII

S. No	Course Code	Course Name	L	T	P	C
1	CH516T	Conceptsin organic synthesis	3	0	0	6
2		Program Elective V/ Minor Project	2	1	0	6
3		HEART Elective-I	3	0	0	6
5		Institute Elective – III	2	1	0	6
6	CH501L	Chemistry laboratory-IV	0	0	3	3
	Total credit					**27**

Semester VIII						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CH401P	Program Elective VI/ Minor Project	2	1	0	6
2		HEART Elective-II	3	0	0	6
3		Institute Elective -IV	2	1	0	6
4		Institute Elective – V	2	1	0	6
	Total credit					**24**

Semester IX & X			
1		Independent Project	30

Semester IX & X			
1		Independent Project	30
Overal Credits Required (Minimum)			**325**

BSMS- Biology

SEMESTER - I						
S.No	**C. Code**	**Course**	**L**	**T**	**P**	**C**
1	MA108T	Calculus	3	1.5	0	9
2	PH101T	Quantum Physics and Applications	2	1	0	6
3	CH101T	Fundamental Concepts and Applications of Chemistry	3	0	0	6
4	BB101T	Introduction to Modern biology	3	0	0	6
5	PH101L	Hands on Science Laboratory - I	0	0	3	3
6	CS101C	Computer Programming	3	0	3	9
7	HS101T	Design Thinking and Creativity	1	0	0	PP/NP
8	CC	NSO/NSS/NCC/NCA	0	0	2	2
	First Semester Total Credits					41

Semester - II						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CS101T	Data Structures and Algorithms	3	0	0	6
2	BB201T	Biomolecules	2	1	0	6
3	MA109T	Linear Algebra and Differential Equations	3	1.5	0	9
4	PH102T	Electricity and Magnetism	2	1	0	6
5	CH203T	States of Matter	3	0	0	3
6	CH201T	Organic Chemisrty	3	0	0	3
7	CS101L	Data Structures and Algorithms Laboratory	0	0	3	3
8	PH101L	Hands on Science Laboratory - 1	0	0	3	3
9	CC	NSO/NSS/NCC/NCA	0	0	2	2
		Total Credits				41

Semester III						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	HS201T	Economics	3	0	0	6
2	BB401T	Basics of Cell Biology and Genetics	3	0	0	6
3	CH204T	Physical Organic and Bioorganic Chemistry (1st Half)	3	0	0	3
4	CH202T	Inorganic Chemistry (2nd Half)	3	0	0	3
5	MA401T	Introduction to Probability Theory	3	1	0	8
6	PH201T	Waves, Oscillations & Optics	2	1	0	6
7	MA102L	Mathematics Laboratory	0	0	3	3
		Total Credits				35

Semester IV						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	BB402T	Metabolism and Bioenergetics	2	1	0	6
2	BB403T	Biophysics (1st Half)	3	0	0	3
3	BB201L	Biology Lab I	0	0	3	3
4	BB406T	Biophysical Methods	3	0	0	3
5		Institute Elective – I	3	0	0	6
6		Program Elective-I	3	0	0	6
7		HSS Elective-I	3	0	0	6
		Total Credits				33

Semester V						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	BB405T	Molecular Biology	2	1	0	6
2	BB502T	Genomics and proteomics (1st Half)	3	0	0	3
3	BB501T	Biostatistics (2nd Half)	3	0	0	3
4	BB202L	Biology Lab II	0	0	6	6
5		Institute Elective – II	3	0	0	6
6		Program Electives				6
		Total Credits				30

Semester VI						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	CE301T	Environmental Studies	3	0	0	6
2	BB402L	Bioinformatics Lab	0	0	3	3
3		Program Elective	2	1	0	6
4		Institute Elective – III				6
5	BB301L	Biology Lab III	0	0	6	6
		Total Credits				27

Semester VII						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	BB602T	Immunology	2	1	0	6
2	BB401L	Biology Lab IV	0	0	6	6
3		Program Elective	3	0	0	6
4		Institute Elective -IV	3	0	0	6
5		HSS Elective-II	3	0	0	6
		Total Credits				30

Semester VIII						
S. No	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1		Program Elective	3	0	0	6
2		Institute Elective -V	3	0	0	6
3		Minor Project				12
		Total Credits				24

Semester IX & X			
1		Independent Project	60
		Total Credits	60
Overall Credits Required (Minimum)			**321**